



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

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ASSIGNMENT No: 9

TITLE: Write a Java program to demonstrate the use of the final keyword with variables, methods, and classes.

Problem Statement

Write a Java program to demonstrate the use of the final keyword with variables, methods, and classes.

Learning Objectives

To implement the final keyword with variables, methods, and classes.

Theory

The final keyword restricts modification in Java programs. A final variable cannot be reassigned after initialization. A final method cannot be overridden by subclasses, while a final class cannot be inherited. This improves security, prevents accidental modifications, and increases program reliability. Final is commonly used for constants and immutable classes.

Outcome:

The final keyword was successfully applied to a variable, a method, and a class, demonstrating that final variables cannot be reassigned, final methods cannot be overridden, and final classes cannot be inherited.

```
class Constants {
    final double PI = 3.14159;

    void showPI() {
        System.out.println("Value of PI: " + PI);
    }
}

class Vehicle_ {
    final void showType() {
        System.out.println("This is a generic vehicle");
    }
}

class Car extends Vehicle_ {

    void showBrand() {
        System.out.println("This is a Car");
    }
}

final class Utility {
    void printMessage() {
        System.out.println("Utility class method called");
    }
}

public class JAVA_EXP_9 {
    public static void main(String[] args) {

        // 1. final variable
        Constants c = new Constants();
        c.showPI();
        // c.PI = 3.15;    // Not allowed - compile-time error, PI is final

        System.out.println();

        // 2. final method
        Car car = new Car();
        car.showType();
        car.showBrand();
    }
}
```

```
        System.out.println();

        // 3. final class
        Utility u = new Utility();
        u.printMessage();
    }
}
```

Outcome:

```
C:\Users\Aashray\OneDrive\Desktop\JAVA_PRACTICALS>java JAVA_EXP_9.java
Picked up JAVA_TOOL_OPTIONS: -Dstdout.encoding=UTF-8 -Dstderr.encoding=UTF-8
Value of PI: 3.14159

This is a generic vehicle
This is a Car

Utility class method called
```

Conclusion:

Therefore, for the above experiment we got to learn the use of final keyword with variables, methods and classes.

Exercise

1. Create a Bank Account program where account number is final and cannot be changed.

Source Code:

```
1  class BankAccount {
2
3      final int accountNumber;    // Final variable
4      String accountHolder;
5      double balance;
6
7      BankAccount(int accountNumber, String accountHolder, double balance) {
8          this.accountNumber = accountNumber;
9          this.accountHolder = accountHolder;
10         this.balance = balance;
11     }
12
13     void display() {
14         System.out.println("Account Number : " + accountNumber);
15         System.out.println("Account Holder : " + accountHolder);
16         System.out.println("Balance      : " + balance);
17     }
18
19     Run | Debug | Run main | Debug main
20     public static void main(String[] args) {
21         BankAccount b = new BankAccount(1001, "Aashray", 25000);
22
23         b.display();
24
25         // b.accountNumber = 2002; // Error: Cannot change final variable
26     }
27 }
```

Output:

```
C:\Users\Aashray\Documents\University\3rd sem\JAVA>java assignment8_a.java
Picked up JAVA_TOOL_OPTIONS: -Dstdout.encoding=UTF-8 -Dstderr.encoding=UTF-8
Account Number : 1001
Account Holder : Aashray
Balance      : 25000.0
```

2. Create a Library Book Management program where the book ISBN is declared as final and cannot be changed once assigned. Display the book's ISBN, title, author, and price.

Source Code:

```
1  class LibraryBook {
2
3      final String isbn;    // Final variable
4      String title;
5      String author;
6      double price;
7
8      LibraryBook(String isbn, String title, String author, double price) {
9          this.isbn = isbn;
10         this.title = title;
11         this.author = author;
12         this.price = price;
13     }
14
15     void display() {
16         System.out.println("ISBN   : " + isbn);
17         System.out.println("Title  : " + title);
18         System.out.println("Author : " + author);
19         System.out.println("Price : " + price);
20     }
21
22     Run | Debug | Run main | Debug main
23     public static void main(String[] args) {
24
25         LibraryBook book = new LibraryBook(
26             isbn: "978-81-123456-7",
27             title: "alchamiest",
28             author: "Paulo Coelo",
29             price: 599.0);
30
31         book.display();
32
33         // book.isbn = "12345"; // Error: Cannot change final variable
34     }
35 }
```

Output:

```
C:\Users\Aashray\Documents\University\3rd sem\JAVA>java assignment9_b.java
Picked up JAVA_TOOL_OPTIONS: -Dstdout.encoding=UTF-8 -Dstderr.encoding=UTF-8
ISBN   : 978-81-123456-7
Title  : alchamiest
Author : Paulo Coelo
Price  : 599.0
```